

Multi-Agent Cooperation for Computing Power Scheduling in UAVs Empowered Aerial Computing Systems

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Abstract—In the paradigm of ubiquitous edge computing, with those advantages, e.g., high mobility, fast response, flexibility and controllability, and low cost of use, Unmanned Aerial Vehicles (UAVs) could be used not only as relays to assist with data collection, but also as computing power nodes to process uncomplicated computational workloads from ground users. Especially, UAVs could be employed to provide alternative computing power resources in field, lake, post-disaster and other complex regional environments. In this paper, to address the issue of computing power scheduling in UAVs empowered aerial computing systems, a scenario where multiple UAVs from the same departure station cooperatively fly over hovering points and achieve the data collection and computation in a decentralized manner is investigated. Nevertheless, due to limited onboard battery capacities of UAVs and diverse service requests of ground users, it is necessary to optimize energy efficiency and service fairness for improving mission execution capabilities of UAVs and the quality of service (QoS) experienced by ground users, and a joint optimization problem of energy efficiency and service fairness is formulated. Through considering complex coupling associations among the departure station, flight paths and hovering points of UAVs, the problem is investigated from the trajectory planning of UAVs and the location planning for both the departure station and hovering points. Proving investigations to be Markov decision processes (MDP), multi-agent cooperation approaches are proposed as promising solutions, and simulation results have been shown to demonstrate that the performance

achieved by the proposal outperforms that achieved by schemes commonly used in literatures.

Index Terms—Aerial computing systems, computing power scheduling, multi-agent cooperation, energy efficiency, service fairness.

I. INTRODUCTION

WITH the emergence of mobile applications, those compute-intensive and delay-sensitive intelligent services impose great pressures on energy and computing power of mobile terminals, and diverse service requests could not be well satisfied with the limited resource capacities. To address these challenges, being different from the centralized processing method adopted by the traditional cloud computing, mobile edge computing (MEC) as an emerging distributed network structure could sink the computing tasks of mobile terminals to the network edge, which effectively shorten the transmission delay of computing data and relieve the computing pressure of the core network, so as to meet the high-reliability and low-delay computing requests of mobile terminals, and greatly improve the computing capability of the network [1]. Nevertheless, due to the difficulty of deploying edge computing related infrastructures in fixed locations in some complex environments, e.g., field, lake, and post-disaster, mobile edge computing has been an effective alternative solution [2].

In the last decade, through being integrated into wireless sensing networks for Internet of Things (IoT) applications of air-ground collaboration, UAVs are popular to be used as relays to assist with the sensing data collection, and mobile access points to carry out emergency rescue [3], [4]. Nowadays, promoted by the enhancement of on-chip computing power, UAVs also could be taken as computing power nodes to process uncomplicated computational workloads from ground users [5], [6]. Therefore, in the paradigm of ubiquitous edge computing, with the advantages of high mobility, fast response, flexibility and controllability, and low cost of use, UAVs assisted aerial computing systems could break through the limitations of traditional edge computing applications in terms of inflexible deployment, constrained geographical location and coverage capabilities, and could greatly expand the application of edge computing through improving the network computing capability and the QoS experienced by ground users.

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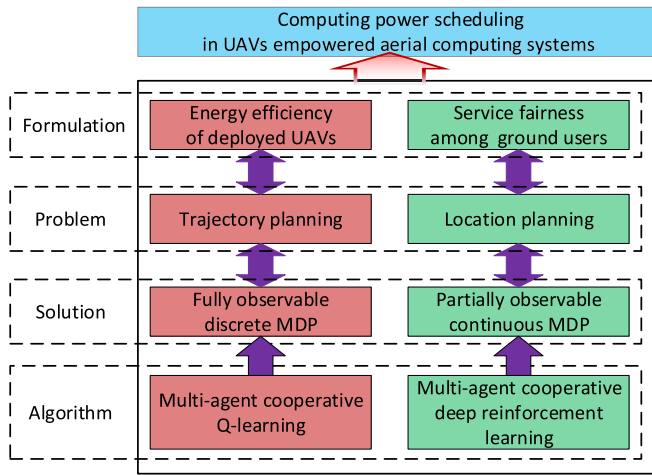


Fig. 1. The diagram of investigations in this paper.

Being different from terrestrial edge computing power nodes deployed in fixed locations, UAVs as mobile computing power nodes could fly over the target area and cooperatively achieve the data collection and computation in a decentralized manner. Nevertheless, due to limited onboard battery capacities embedded in UAVs and diverse service requests put forwarded by ground users, operating the UAVs assisted aerial computing systems in optimal fashion still remains a challenging issue that is necessary to be reasonably solved [7]. To this end, to address the issue of computing power scheduling in UAVs empowered aerial computing systems, a scenario where multiple UAVs fly from the same departure station to achieve the cooperation is investigated in this paper. Additionally, since the cooperation of flight paths of UAVs is directly influenced by the departure station, and the efficiency of data collection and computation is directly influenced by flight paths and hovering points, the departure station, flight paths and hovering points of UAVs have obvious and complex coupling associations, the fundamental challenge is to determine the optimal departure station, flight paths and hovering points. To address these challenges, with the main contributions summarized as follows, a joint optimization of energy efficiency and service fairness is investigated in this paper to improve mission execution capabilities of UAVs and the service experience of ground users. The diagram of investigations is shown in Fig. 1.

- Being different from existing proposals, both data collection and computation are investigated in multi-UAV assisted aerial computing systems, and with the consideration of limited onboard battery capacity of UAV and diverse service requests of ground users, both energy efficiency and service fairness are jointly optimized for computing power scheduling.
- For improving mission execution capabilities of UAVs and the QoS experienced by ground users, achieving the energy efficiency is formulated as jointly minimizing the energy utilization and the standard deviation of energy consumption, and achieving the service fairness is formulated as jointly minimizing the maximum in minimum hop counts between ground users and their candidate serving UAVs, and the standard deviation of hop counts.

- Through fully considering complex coupling associations among the departure station, flight paths and hovering points of UAVs, the trajectory planning of UAVs is investigated to maximize the energy efficiency, which is proven to be a fully observable discrete MDP, and an approach of multi-agent cooperative Q-learning is proposed as a promising solution.
- Also, the location planning for the departure station and hovering points of UAVs is investigated to maximize the service fairness among ground users, which is proven to be a partially observable continuous MDP, an approach of policy optimization based multi-agent cooperative deep reinforcement learning within the architecture of Actor-Critic is proposed as a promising solution.

The rest of this paper is organized as follows. In Section II, a brief review of UAV applications is firstly stated, on the basis of the analysis, research motivations of this paper are discussed. In Section III, basic definitions in the considered scenario are clearly given. In Section IV, formulations of investigated problems including optimization objectives and constraints are comprehensively discussed. In Section V, through proving investigated problems to be Markov decision processes, multi-agent cooperative approaches are proposed as promising solutions. In Section VI, experiments and analysis are addressed to demonstrate the effectiveness of investigations. In Section VII, the summarization and future work are discussed.

II. RELATED WORK AND MOTIVATIONS

Benefiting from advantages of high mobility, fast response, flexibility and controllability, and low cost of use, UAVs have been widely employed in various fields, e.g., intelligent agriculture, intelligent transportation, disaster management, environment monitoring and construction of smart city, through acting as mobile access points for data collection and mobile computing nodes for data computation. In the meantime, the involved theoretical and technical challenges also have attracted extensive attention from academia and industry, and some research progress has been obtained in some specific applications with different considerations.

A. UAV-Assisted Data Collection

With high mobility and wide coverage, UAVs have providing new opportunities and conveniences for effective and timely data collection. Through reviewing application scenarios and key technologies, Wei et al., discussed the mathematical model of UAV assisted data collection, the mode of UAV assisted data collection and some open problems [8]. To minimize the energy consumption and completion time while achieving reliable data collection, through using a general fading channel model for sensor-UAV links, Zhan et al., investigated the trajectory planning and sensor wake-up schedule [9]. Yuan et al., also proposed a joint scheme of UAV trajectory and user scheduling to achieve the completion time minimization of data collection [10].

Through considering the sensing data collection with an UAV on a straight line, Gong et al., jointly optimized data

collection intervals, the UAV's speed, and sensors' transmission powers to achieve the minimization of UAV flight time [11]. With the UAV flight control, Li et al., also considered the limited energy of UAV to achieve the fairness of data collection in the scenario that the UAV could not achieve direct data collection for all devices [12]. Through designing fully or partially data collection problems, a scheme of maximizing the accumulative volume of data collection with energy-constrained UAV was also investigated in [13]. With the consideration of energy-constrained IoT devices and target data upload deadlines, Samir et al., formulated the joint optimization of UAV trajectory and radio resource allocation as a mixed integer non-convex problem to maximize the number of served IoT devices [14]. While considering the UAVs flying range, communication range and energy constraints, a scheme was also proposed in [15] to improve the network resource allocation fairness and efficiency.

To maintain the freshness of data collection, through comprehensively considering UAV trajectory and time for energy harvesting, Hu et al., designed a scheme to minimize the average Age of Information (AoI) [16]. With the consideration that the total AoI depends on the UAV flight time and the data collection time at hovering points, Zhu et al., designed a joint optimization problem of selecting hovering points and determining the visiting order to hovering points [17]. Gao et al., also jointly optimized the sensor uploading time, the UAV flight time and the data offloading time to improve the information freshness [18].

However, those proposals in literatures focused on simplified 2-D scenarios with perfect channel state information and generally formulated the challenges as non-convex problems solved by heuristic algorithms. Unlike those proposals, with the consideration of practical 3-D scenarios with imperfect channel state information and the usage of deep reinforcement learning approach, Wang et al., investigated the optimization of UAV trajectory to minimize the completion time of data collection [19], Nguyen et al., optimized the UAV trajectory planning while maximizing the collected data volume to enhance the throughput [20], Chen et al., jointly optimized the UAV trajectory and user association to maximize the network throughput and energy efficiency [21], Fan et al., also jointly optimized the UAV trajectory, sensors scheduling and discrete phase shifts of the reconfigurable intelligent surface (RIS) to minimize the AoI and maintain the data freshness [22].

B. UAV-Assisted Mobile Edge Computing

MEC as an emerging paradigm has been demonstrated to promise low latency and low energy services. In extreme scenarios where terrestrial infrastructures are unavailable, with a certain on-chip computing power, UAV acting as a mobile edge computing node, could be taken as an alternative [23].

Through jointly optimizing the computing power, user associations and ground user locations, Shah et al., presented a maximization of UAV deployment and computation efficiency [24]. Taking the temporarily passing UAV as the relay and edge computing node to provide computing power service, Yang et al., proposed a learning based channel allocation

and task offloading strategy [25]. To address the joint optimization of queue-based computation offloading and adaptive computing resource allocation, Goudarzi et al., proposed an computing resource allocation model optimized by cooperative evolutionary computation methods [26]. Through considering those constraints, e.g., resources budgets, computation deadline, data compression ratio and UAVs trajectory limitations, Cheng et al., investigated an offloading scheme to improve the overall computing performance and energy efficiency [27].

In UAV-mounted MEC systems, to assume that priorities of computation tasks from different users are time-varying and incorporated, Zhou et al., proposed a priority-aware resource scheduling problem which was solved by the deep Q-learning algorithm optimizing the UAV hotspot selection and task offloading [28]. For avoiding the problem of overloading and congestion at a UAV while achieving the minimization of makespan, Hoa et al., presented a cooperative multi-hop offloading problem solved by a deep reinforcement learning approach [29]. With the consideration of long-term constraints on queue stability and computational delay, Hoang et al., proposed an online resource allocation strategy, and used deep reinforcement learning approaches to achieve the minimization of overall energy consumption [30].

C. Motivations

From above reviews, UAV-assisted data collection and computation as an on-demand decision making problem is NP-hard that could be solved by various optimization methods with different effectiveness. With the acknowledgement that those proposals in literatures are effective in some specific scenarios with different considerations, however, there is still lack of complete considerations that are necessary to be conducted further improvements. Some proposals only employed a single UAV to provide data collection and mobile computing service, while it was inadequate to take the cooperation among UAVs into consideration in those proposals employing multiple UAVs as well. In most proposals, the UAV trajectory planning was a general method used to optimize the energy consumption and service latency, and the departure station was generally deployed at a fixed location. In fact, the initial location of departure station in the considered scenario would have a great impact on conducting the subsequent trajectory planning of UAVs. However, those proposals were lack of complete consideration of complex coupling associations among the departure station, flight paths and hovering points of UAVs. Additionally, there has been sufficient research progress on UAVs acting as multiple access points for data collection, and UAV as edge computing node in some typical applications also has been investigated to a certain extent, however, for the computing power scheduling in UAVs empowered aerial computing systems, the involved theoretical and technical challenges still have not attracted extensive attention.

Motivated by the above analysis of existing proposals, but being different from them, a joint optimization of energy efficiency and service fairness in UAVs empowered aerial computing systems is investigated in this paper, which fully considers the cooperation among multiple UAVs and complex coupling associations among the departure station, flight paths

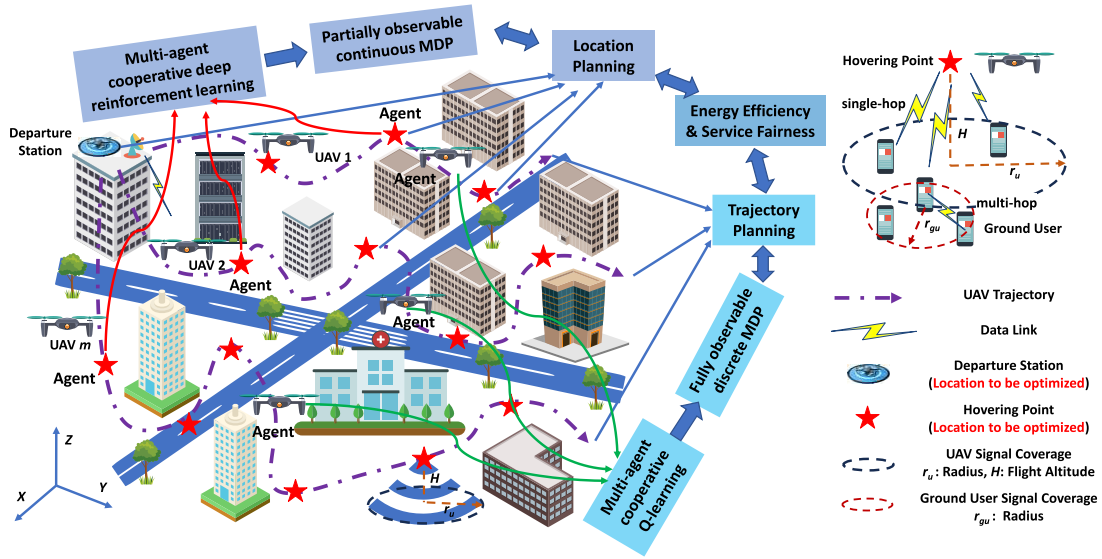


Fig. 2. Multi-UAV empowered edge computing power network.

TABLE I
BRIEF INTRODUCTIONS OF INVOLVED SYMBOLS AND CORRESPONDING EXPLANATIONS

Symbols	Explanations
$GU = \{gu_1, \dots, gu_N\}$	Ground users.
$D = \{D_1, \dots, D_N\}$	Data volumes generated by distinct ground users.
$UAV = \{u_1, \dots, u_M\}$	UAVs to be deployed.
$C = \{c_1, \dots, c_M\}$	Computing capacities of distinct UAVs.
$S(o) = \{s_0\}$	Departure station whose location is to be optimized.
$S(h) = \{s_1, \dots, s_K\}$	Hovering points of UAVs whose locations are to be optimized.
$\mathcal{M}(hop)$	Matrix of hops between entities (ground users, hovering points).
x_{ij}^m	A binary variable indicating whether u_m flies between the two hovering points, s_i and s_j .
y_k^n	A binary variable indicating whether the data of gu_n is processed by the UAV hovering at s_k .
$GU_k = \{gu_k^1, \dots, gu_k^{N_k}\}$	Ground users served by u_m hovering at s_k .
D_k	Data volume to be processed by u_m hovering at s_k .
E_m^{fly}	Energy consumption of flight caused at u_m .
E_m^{col}	Energy consumption of data collection caused at u_m .
E_m^{com}	Energy consumption of data computation caused at u_m .
E_m^{hover}	Energy consumption of hovering caused at u_m .

and hovering points, and conducts investigations from the trajectory planning of UAVs and the location planning for both the departure station and hovering points of UAVs to achieve computing power scheduling.

III. DEFINITIONS

As shown in Fig. 2, in the considered scenario with a regional scale of $\mathbb{R}^3$, the involved parameters could be defined as follows, and brief introductions of involved symbols and corresponding explanations are stated in Table I.

Ground users represented as $GU = \{gu_1, \dots, gu_n, \dots, gu_N\}$ act the random location distribution, and the data volume generated by distinct ground users is represented as $D = \{D_1, \dots, D_n, \dots, D_N\}$. M UAVs represented as $UAV = \{u_1, \dots, u_m, \dots, u_M\}$ whose embedded computing capacities are heterogeneous and represented as $C = \{c_1, \dots, c_m, \dots, c_M\}$, where, c_m (unit: cycles per second)

represents the computing capacity of u_m and $c_m > 0$. A departure station whose location is to be optimized, and could be assumed as $S(o) = \{s_0\}$. Hovering points of UAVs is represented as $S(h) = \{s_1, \dots, s_k, \dots, s_K\}$ to be optimized as well. Accordingly, ground users, the departure station and hovering points of UAVs could constitute an undirected graph, and the UAVs assisted edge computing power network could be represented as $G(V, E)$, in which, V represents the set of locations of ground users, the departure station and hovering points of UAVs, and E represents the set of edges among ground users, edges between ground users and hovering points and edges among hovering points.

For establishing a reasonable data collection and computation model, the following definitions are necessary to be stated.

Definition 1 (Neighbor): In $G(V, E)$, assuming that the communication radius of terminals owned by ground users is represented as r_{gu} , the communication radius of UAVs is

represented as r_u and the flight altitude is H , for any two nodes within the area, v_i and v_j , if the Euclidean distance $d(v_i, v_j)$ could satisfy one of the following conditions, v_i and v_j are neighbors.

- $(v_i, v_j) \in GU$ & $d(v_i, v_j) \leq r_{gu}$.
- $(v_i, v_j) \in GU \times S(h)$ & $d(v_i, v_j) \leq \sqrt{r_u^2 - H^2}$.

Definition 2 (Connex): Assuming that $\Phi = v_0 e_1 v_1 e_2 \dots e_\tau v_\tau$ is a sequence of alternating points and edges, for any i in the range of $[1, \tau]$, if the two adjacent nodes v_{i-1} and v_i are neighbors, Φ has the connectivity.

Definition 3 (Reachability): For any two nodes v_i and v_j , if there is a *Connex*, v_i and v_j have the reachability.

Definition 4 (Hop-Count): For any two nodes v_i and v_j , if there is a *Reachability* and the number of edges connecting v_i and v_j is $N(e)$, the hop count could be represented as $R(v_i, v_j) = N(e)$, otherwise, $R(v_i, v_j) = \infty$. Accordingly, in terms of locations of ground users and hovering points, using the Floyd algorithm to determine the shortest path between nodes, the hop count of each pair of ground users and that of each pair of associated ground user and hovering point could be calculated and constitute a matrix, $\mathcal{M}(hop)$.

The two binary variables x_{ij}^m and y_k^n are also introduced to state associations among UAVs, hovering points and ground users. x_{ij}^m indicates that whether u_m flies from the hovering point s_i to another hovering point s_j , in that case and $i \neq j$, $x_{ij}^m = 1$, otherwise, $x_{ij}^m = 0$. y_k^n indicates that whether the data of gu_n is collected and calculated by the UAV hovering at s_k , in that case, $y_k^n = 1$, otherwise, $y_k^n = 0$.

If the data generated by a set of ground users, $GU_k = \{gu_k^1, gu_k^2, \dots, gu_k^{N_k}\}$, would be collected and calculated by u_m hovering at s_k , where, N_k represents the number of ground users in GU_k and $N_k = \sum_{n=1}^N y_k^n$, the data volume to be

processed by u_m could be represented as $D_k = \sum_{i=1}^{N_k} D_k^i$, which

could be further defined as $D_k = \sum_{n=1}^N y_k^n \cdot D_n$.

In the considered scenario, since different ground users served by the same UAV would use the same spectrum resource at the same time, the technology of Orthogonal Frequency Division Multiple Access (OFDMA) is employed in the uplink to avoid the conflict [31], [32], where the frequency bandwidth B_m of u_m could be evenly divided into N_k equal sub-bands to provide the service for a maximum of N_k ground users. With the Shannon formula, the uplink data transmission rate for gu_k^i could be defined as

$$r_k^i = \frac{B_m}{N_k} \cdot \log \left(1 + \frac{P_k^i \cdot g_k^i}{\frac{B_m}{N_k} \cdot \theta} \right), \quad (1)$$

where P_k^i is the uplink transmission power of gu_k^i , g_k^i is the uplink channel gain between gu_k^i and u_m , and θ is the noise spectral power density.

On the basis of above, the following definitions could be further introduced.

Definition 5 (Energy Consumption of Fight): Assuming that δ represents the energy consumption coefficient of UAV

flight, for u_m , the energy consumption of flight could be defined as

$$E_m^{fly} = \delta \cdot \sum_{(i,j) \in \{S(o), S(h)\}^2} x_{ij}^m \cdot d_{ij}, \quad (2)$$

where d_{ij} is the Euclidean distance between the two hovering points, s_i and s_j .

Definition 6 (Energy Consumption of Data Collection):

In terms of above definitions, the transmission delay of data generated by gu_k^i could be represented as $t_k^i(up) = D_k^i / r_k^i$. Accordingly, the energy consumption of data collection for gu_k^i could be defined as $E_k^i(up) = \int_0^{t_k^i(up)} P_k^i(t) dt$, and the total energy consumption of data collection of u_m hovering at s_k could be further defined as

$$E_m^{col} = \sum_{i=1}^{N_k} \int_0^{t_k^i(up)} P_k^i(t) dt. \quad (3)$$

Definition 7 (Energy Consumption of Data Computation):

The energy consumption of data computation of u_m hovering at s_k could be defined as

$$E_m^{com} = \kappa_m \cdot c_m^2 \cdot D_k, \quad (4)$$

where κ_m represents the energy coefficient determined by the chip structure of the UAV.

Definition 8 (Energy Consumption of Hovering): The time of data collection and that of data computation of u_m hovering at s_k respectively could be represented as $t_k(up) = \max_{i=1,2,\dots,N_k} \{t_k^i(up)\}$ and $t_k(exe) = D_k / c_m$. Assuming that ξ is the energy consumption coefficient of UAV hovering, the total energy consumption of hovering therefore could be defined as

$$E_m^{hover} = \xi \cdot \left(\max_{i=1,2,\dots,N_k} \{t_k^i(up)\} + \frac{D_k}{c_m} \right). \quad (5)$$

IV. FORMULATIONS

In the considered scenario, due to limited onboard battery capacities of UAVs and diverse service requests of ground users, it is necessary to optimize the trajectory planning of UAVs and the location planning for both the departure station and hovering points of UAVs, for improving the energy efficiency of UAVs and the service fairness experienced by ground users. In terms of above definitions, the following two optimization objectives could be formulated.

A. Energy Efficiency

Through considering the timeliness of data collection and computation, and limited onboard battery capacities of UAVs for satisfying necessary energy supplies of making a return trip while completing data collection and calculation, to provide sufficient services for served ground users with limited energy consumption and achieved load balance, the first optimization objective is introduced to minimize the energy utilization and the standard deviation of energy consumption through conducting reasonable trajectory planning (**TP**) of UAVs, so as

to maximize the energy efficiency of UAVs, which could be defined as

$$\begin{aligned} \min_{\text{TP}} (U(E) + \alpha \cdot STD(E)) \\ = \min_{\text{TP}} \left(\frac{\sum_{m=1}^M \frac{E_m}{E_m^{\text{full}}}}{M} + \alpha \cdot \sqrt{\frac{\sum_{m=1}^M (E_m - \bar{E})^2}{M}} \right), \quad (6) \end{aligned}$$

where $U(E) = \sum_{m=1}^M \frac{E_m}{E_m^{\text{full}}} / M$ represents the average energy utilization of UAVs, $STD(E) = \sqrt{\sum_{m=1}^M (E - \bar{E})^2} / M$ representing the standard deviation of energy consumption of UAVs is an equilibrium coefficient of energy consumption, E_m^{full} represents the full onboard battery capacity of u_m , $E_m = E_m^{\text{fly}} + E_m^{\text{col}} + E_m^{\text{com}} + E_m^{\text{hover}}$ represents the total energy consumption of u_m used to execute a flight mission, and $\bar{E} = \sum_{m=1}^M E_m / M$ represents the average energy consumption of all UAVs.

In general, from the definition of $U(E)$, it is expected to maximize the energy utilization of UAVs with limited onboard battery capacities, however, with a certain number of deployed UAVs in the considered scenario, the minimization setting of energy utilization is adopted here to avoid the problem of UAV purposely executing a flight mission over long distance while increasing the unnecessary energy consumption, and enhance the energy efficiency accordingly. Additionally, in (6), the former optimization objective is used to minimize the energy utilization, so as to eliminate the unnecessary energy consumption of UAVs, the latter is used to minimize the standard deviation of energy consumption, so as to achieve the load balance of UAVs, therefore, a factor α is introduced to balance the relationship between the two optimization objectives, and determine the distinct importance.

B. Service Fairness

The hop count between the ground user and the serving UAV, which indicates the serving delay, has direct influence on the QoS experienced by the ground user. Due to the limited number of UAVs executing flight missions and the limited signal coverage of the UAV with the flight altitude H , some ground users to be served by a UAV would choose other ground users as relay nodes to assist with the data transmission. Nevertheless, onboard battery capacities of terminals owned by ground users are limited as well, the imbalance of hop counts would result in the emergence of ‘energy holes’ around some ground users, which would break up the network topology with the loss of connectivity and consequently shorten the network lifespan, as a result, those ground users could not experience the adequate service. Accordingly, the second optimization objective is introduced to minimize the maximum in minimum hop counts between ground users and their candidate serving UAVs and the standard deviation of hop counts through conducting reasonable location planning (LP) for both the departure station and hovering points, so as to maximize the service fairness among ground users, which

could be formulated as

$$\begin{aligned} \min_{\text{LP}} \left(\max_{n=1, \dots, N} R_n + \beta \cdot STD(R) \right) \\ = \min_{\text{LP}} \left(\max_{n=1, \dots, N} \left(\min_{k=0, \dots, K} R(gu_n, s_k) \right) + \beta \cdot \sqrt{\frac{\sum_{n=1}^N (R_n - \bar{R})^2}{N}} \right), \quad (7) \end{aligned}$$

where $R_n = \min_{k=0, \dots, K} R(gu_n, s_k)$ represents the minimum of hop counts between gu_n and the chosen serving UAV hovering at s_k , $\bar{R} = \sum_{n=1}^N R_n / N$ represents the average of minimum hop counts, $STD(R) = \sqrt{\sum_{n=1}^N (R_n - \bar{R})^2} / N$ representing the standard deviation of minimum hop counts, is introduced as a coefficient of fairness. The greater the standard deviation, the worse the fairness of different ground users in requesting computing power. Additionally, in (7), the former optimization objective is used to minimize the hop counts between any one ground user and the chosen serving UAV, so as to enhance the energy efficiency of data transmission, the later is used to minimize the standard deviation of hop counts, so as to balance the energy consumption of ground users, therefore, a factor β is introduced to balance the relationship between the two optimization objectives, and determine the distinct importance.

C. Constrains

In the considered scenario, taking into account of the air-ground limitation, the problem of recharge and recondition of UAVs, the association between ground user and UAV, the full coverage with limited UAVs, safe distances of UAVs flight and limited onboard battery capacities of UAVs, the following constraints are necessary to be introduced to solve the two formulated optimization objectives in an optimal fashion.

Constraint 1 (Area): Taking into account of the air-ground limitation, locations of ground users, the location of departure station (to be optimized) and locations of hovering points of UAVs (to be optimized) should be in the air-ground of $\mathbb{R}^3$. Meanwhile, due to the consideration of relay relationship among ground users, to ensure the isolation of service areas covered by hovering points, there should be a certain distance between hovering points. Therefore, *Constraint 1* could be defined as

$$\begin{cases} d(s_i, s_j) > \bar{h}, \forall (s_i, s_j) \in S(h)^2, s_i \neq s_j, \\ GU, S(o), S(h) \subseteq \mathbb{R}^3, \end{cases} \quad (8)$$

where, $\bar{h}$ is the threshold of distance.

Constraint 2 (Number of Flights): Through considering recharge and recondition of UAVs, each UAV only make up to one return trip, and *Constraint 2* could be defined as

$$\sum_{k=1}^K x_{s_0 s_k}^m \leq 1, \forall m \in \{1, 2, \dots, M\}. \quad (9)$$

Constraint 3 (Association): To ensure that all ground users could be fully served, and the integrity of data collection and computation for ground users, each ground user could only be associated with a hovering point of UAV, and

Constraint 3 could be defined as

$$\sum_{k=0}^K y_k^n = 1, \forall n \in \{1, 2, \dots, N\}. \quad (10)$$

Constraint 4 (Full Coverage): With the limited number of deployed UAVs, to ensure each ground user to be served by only one UAV, the serving coverage area of each determined hovering point should be visited by only one UAV, and *Constraint 4* could be defined as

$$\begin{cases} \sum_{m=1}^M \sum_{k=1}^K x_{s_0 s_k}^m \leq M, \\ \sum_{m=1}^M x_{s_i s_k}^m = \sum_{m=1}^M x_{s_k s_j}^m = 1, \\ \forall s_k \in S(h), (s_i, s_j) \in \{S(o), S(h)\}^2, k \neq i, k \neq j. \end{cases} \quad (11)$$

Constraint 5 (Flight Path): To effectively manage deployed UAVs and avoid collisions with safe flight distance, after each UAV completing the data collection and computation at hovering points along the determined flight path, each UAV follows its original path back to the departure station, and *Constraint 5* could be defined as

$$\sum_{(i,j) \in \{S(o), S(h)\}^2} x_{ij}^m = \sum_{(j,i) \in \{S(o), S(h)\}^2} x_{ji}^m, i < j. \quad (12)$$

Constraint 6 (Energy Consumption): Assuming that each UAV is not rechargeable during the flight, the maximum energy consumption used to execute a flight mission must not be exceed the full onboard battery capacity denoted as E_m^{full} , and *Constraint 6* could be defined as

$$\max_{m \in \{1, \dots, M\}} E_m \leq E_m^{full}. \quad (13)$$

D. Joint Optimization Problem

In terms of above defined formulations for optimizing the energy efficiency of UAVs and service fairness among ground users, and introduced constraints, the joint optimization problem for achieving reasonable computing power scheduling in the considered scenario could be stated as

$$\begin{aligned} & \begin{cases} \mathcal{P1} : \min_{\text{TP}} (U(E) + \alpha \cdot \text{STD}(E)), \\ \mathcal{P2} : \min_{\text{LP}} \left(\max_{n=1, \dots, N} R_n + \beta \cdot \text{STD}(R) \right), \end{cases} \\ \text{s.t. } & C1 : \begin{cases} d(s_i, s_j) > \bar{h}, \forall (s_i, s_j) \in S(h)^2, s_i \neq s_j, \\ GU, S(o), S(h) \subseteq \mathbb{R}^3, \end{cases} \\ & C2 : \sum_{k=1}^K x_{s_0 s_k}^m \leq 1, \forall m \in \{1, 2, \dots, M\}, \\ & C3 : \sum_{k=0}^K y_k^n = 1, \forall n \in \{1, 2, \dots, N\}, \\ & C4 : \begin{cases} \sum_{m=1}^M \sum_{k=1}^K x_{s_0 s_k}^m \leq M, \\ \sum_{m=1}^M x_{s_i s_k}^m = \sum_{m=1}^M x_{s_k s_j}^m = 1, \\ \forall s_k \in S(h), (s_i, s_j) \in \{S(o), S(h)\}^2, \\ k \neq i, k \neq j, \end{cases} \end{aligned}$$

$$\begin{aligned} C5 : & \sum_{(i,j) \in \{S(o), S(h)\}^2} x_{ij}^m = \sum_{(j,i) \in \{S(o), S(h)\}^2} x_{ji}^m, i < j, \\ C6 : & \max_{m \in \{1, \dots, M\}} E_m \leq E_m^{full}. \end{aligned} \quad (14)$$

V. SOLUTIONS

A. Multi-Agent Cooperation for Energy Efficiency

Through fully considering complex coupling associations among the departure station, flight paths and hovering points of UAVs, the trajectory planning of UAVs is investigated to maximize the energy efficiency, which could be transformed into a discrete MDP that is fully observable for multiple agents.

Proof: For the problem of trajectory planning, the involved agents are M UAVs, which could be represented as $\Upsilon_u \triangleq \{u_1, u_2, \dots, u_M\}$. To execute a flight mission, starting from the departure station s_0 , an agent would make a return trip through traversing a finite number of discrete hovering points $\{s_k(x_k, y_k, z_k)\}_{k \in \{0, \dots, K\}}$, that is, $S(o) \rightarrow S(h) \rightarrow S(o)$. For determining traversed hovering points, in terms of the state of the agent currently hovering at s_k , u_m would perform an action a_k according to the discrete sequence decision mode, and select the next hovering point s_{k+1} , $s_{k+1} \in S(h) \cup S(o)$. Therefore, the decision-making process could be considered as that the choice of hovering point is related to the state of the agent at the current hovering point, but has nothing to do with previous states, that is, $p(s_{k+1} | (s_k, a_k), \dots, (s_0, a_0)) = p(s_{k+1} | (s_k, a_k))$. To improve the efficiency of trajectory planning while satisfying defined constraints, agents communicate with each other to share the observed information, $\mathcal{O}_{u_m}$, including traversed hovering points and energy consumption, therefore, the environment is fully observable for trajectory planning. In conclusion, the problem of trajectory planning could be transformed into a fully observable discrete MDP. ■

Hence, a tuple $\langle \mathcal{S}_u, \mathcal{A}_u, \mathcal{O}_u, \mathcal{R}_u, \mathcal{P}_u, M \rangle$ is defined, $\mathcal{S}_u = \mathcal{S}_{u_1} \times \mathcal{S}_{u_2} \times \dots \times \mathcal{S}_{u_M}$ is the global state space, $\mathcal{A}_u = \mathcal{A}_{u_1} \times \mathcal{A}_{u_2} \times \dots \times \mathcal{A}_{u_M}$ is the action space shared by UAVs, $\mathcal{O}_u = \mathcal{O}_{u_1} \times \mathcal{O}_{u_2} \times \dots \times \mathcal{O}_{u_M}$ is the observation space and $\mathcal{O}_{u_m}$ is generated by u_m local information. $\mathcal{R}_u$ is the reward function, $\mathcal{P}_u : \mathcal{S}_u \times \mathcal{A}_u \rightarrow \mathcal{S}'_u$ is the state transition probability function.

In the considered scenario, to better define the discrete MDP for trajectory planning, the decision-making process could be regarded as a sequencing problem of hovering points for each UAV. Assuming that hovering points (including the departure station) visited by a UAV could constitute a sequence and the maximum sequence length of hovering point for UAV is L , the l -th ($l \in \{1, 2, \dots, L\}$) hovering point of u_m is $u_m(l)$, and $\bigcup_{m=1}^M \bigcup_{l=1}^L u_m(l) = S(o) \cup S(h)$. Additionally, since the number of visited hovering points is different for different UAVs, the maximum sequence length of hovering point sequence is practically different for UAVs, if a UAV has been back to the departure station, the next visiting location would still be the departure station to ensure the consistency of expression. In discrete steps, each agent could interact with the environment to improve the policy of trajectory planning. As an illustration, at step l , u_m could obtain

the current observation $\mathcal{O}_{u_m}(l)$ from the global state space $\mathcal{S}_u(l) \triangleq \{\mathcal{O}_{u_m}(l) | \forall u_m \in \Upsilon_u\}$, $\mathcal{O}_{u_m}(l) : \mathcal{S}_u(l) \rightarrow \mathcal{O}_{u_m}(l)$, subsequently, with the observation, u_m could obtain rewards through executing corresponding actions, and the state could be transferred into a new one $\mathcal{S}_u(l+1)$. The whole process would be repeated until the trained optimal trajectory planning is obtained.

To address the problem of achieving the energy efficiency of UAVs through trajectory planning, detail explanation of each component involved in the discrete MDP is stated as follows. To fully use the advantage of maximizing future rewards [33], [34], a multi-agent cooperative Q-learning algorithm is proposed as a promising solution, which is stated in Algorithm 1.

State: With the formulated optimization problem, the state space has three components, e.g., UAVs, locations of hovering points and energy consumptions of UAVs at hovering points. For u_m hovering at $u_m(l) (\forall l \in \{1, 2, \dots, L\})$, the local state space, $\mathcal{S}_{u_m}(l) \in \mathcal{S}_{u_m}$, could be defined as $\mathcal{S}_{u_m}(l) = \{u_1, \dots, u_M, s_0, \dots, s_K, E_{u_1}(l), \dots, E_{u_M}(l)\}$. The state space shared by agents therefore could be further defined as $\mathcal{S}_u \triangleq \{\mathcal{S}_{u_m}(l)\}$.

Action: For u_m hovering at $u_m(l)$, the chosen next hovering point constitutes the action space, that is, $\mathcal{A}_{u_m} \triangleq \{a_{u_m}(l)\}$, where $a_{u_m}(l) = s_i (s_i \in S(o) \cup S(h))$. Therefore, the global action space shared by agents could be further defined as $\mathcal{A}_u \triangleq \{\mathcal{A}_{u_m}\}$. However, to ensure the serving coverage area of each determined hovering point to be visited by only one UAV, *Constraint 4* should be satisfied in the global action space. Additionally, to ensure all ground users to be fully served by deployed UAVs, and each ground user to be associated with only one hovering point of UAV, if there is any one unserved ground user while all deployed UAVs being on the way departure station, the UAV closest to the unserved ground user would select the action of staying at the current hovering point, and provide service for the unserved ground user through a multi-hop connection, that is, if $a = s_0$, $a = a' \leftarrow a_{u_m}(l-1)$.

Observation: Assuming that agents could communicate with each other to obtain locations of UAVs, currently visited hovering points and energy consumption, for u_m hovering at $u_m(l) (\forall l \in \{1, 2, \dots, L\})$, the local observation space $\mathcal{O}_{u_m}(l) \in \mathcal{O}_{u_m}$, could be defined as $\mathcal{O}_{u_m}(l) = \{u_1, \dots, u_M, s_0, \dots, s_K, E_{u_1}(l), \dots, E_{u_M}(l)\}$, and the global observation space could be further defined as $\mathcal{O}_u \triangleq \{\mathcal{O}_{u_m}(l)\}$.

Reward: To improve the search efficiency of trajectory planning while satisfying defined constraints, and obtain optimal trajectory planning for all UAVs, negative and positive rewards could be introduced from perspectives of the energy consumption caused by each action choice, the full coverage that the serving area of each hovering point could be visited by only one UAV and the next action taken by u_m hovering at $u_m(l)$ should be the choice of hovering point that has not been visited yet, and all UAVs could make a return trip without energy exhaustion.

(1) To enable the path learned by the UAV to be the one with the highest energy efficiency that the UAV could make

Algorithm 1 Multi-Agent Cooperative Q-Learning for $\mathcal{P}1$

Input: Defined parameters, the number of iterations ψ .

Output: Trajectory planning.

```

1: Obtaining the matrix  $\mathcal{M}(hop)$  with Definition 4;
2: for  $s_i \in S(o) \cup S(h)$  do
3:   for  $s_j \in S(o) \cup S(h)$  do
4:     if  $(s_i \neq s_j)$  then
5:       Calculating  $E_{s_i, s_j}^{fly}$  with Definition 5;
6:       Calculating  $E_m(s_j)$  with Definitions 6 ~ 8;
7:       Calculating  $E_{s_i, s_j} = E_{s_i, s_j}^{fly} + E_m(s_j)$ ;
8:     else
9:        $E_{s_i, s_j} = 0$ ;
10:    end if
11:  end for
12: end for
13: Constituting the matrix of energy consumption,  $\mathcal{M}(E)$ ;
14: Initializing a  $Q$  table,  $\forall s \in \mathcal{S}_u$  &  $\forall a \in \mathcal{A}_u$ ;
15: With the number of agents,  $M$ , generating an initial
    solution, and obtaining the value of  $\mathcal{P}1$ ,  $\mathcal{F}(L)$ ;
    /* Updating the  $Q$  table and  $\mathcal{F}(L)$ . */
16: for  $\psi = 1$  to  $\psi_{max}$  do
17:    $\rho \leftarrow 1 - \frac{\psi}{\psi_{max}}$ ;
    /*  $\rho$  is the adaptive exploring rate in Q-learning. */
18:   for  $m = 1$  to  $M$  do
19:     for  $l = 1$  to  $L$  do
20:       Calculating the negative reward  $p_e(l)$ ;
21:       for  $s \in \mathcal{S}_{u_m}(l)$  do
22:         for  $a \in \mathcal{A}_{u_m}$  do
23:           if  $(rand() < \rho)$  then
24:             Selecting  $a$  with  $\mathcal{R}_u(s, a)$  using roulette;
25:           else
26:             Selecting  $a$  with the greatest value of  $Q$ ;
27:           end if
28:           if  $(a = s_0)$  then
29:              $a = a' \leftarrow a_{u_m}(l-1)$ ;
30:             Calculating the negative reward  $p_c(l)$ ;
31:           else
32:              $a = a'$ ;
33:           end if
34:            $Q(s, a) \leftarrow Q(s, a) +$ 
35:              $\lambda \cdot [\mathcal{R}_u(s, a) + \gamma \cdot \max_{a'} Q(s, a') - Q(s, a)]$ ;
36:         end for
37:       end for
38:     end for
39:     if  $(u_m(L+1) = s_0)$  then
40:       Calculating the positive reward  $r(L)$ ;
41:        $Q(s, s_0) \leftarrow Q(s, s_0) +$ 
42:          $\lambda \cdot [\mathcal{R}_u(s, s_0) + \gamma \cdot \max_{s_0} Q(s, s_0) - Q(s, s_0)]$ ;
43:       Obtaining the updated value of  $\mathcal{P}1$ ,  $\mathcal{F}'(L)$ ;
44:       if  $(\mathcal{F}'(L) < \mathcal{F}(L))$  then
45:          $\mathcal{F}(L) \leftarrow \mathcal{F}'(L)$ ;
46:       end if
47:       Determining the path corresponding to  $\mathcal{F}(L)$ ;
48:     end if
49:   end for
50: end for

```

a return trip with enough energy, for each UAV's flight, u_m would be given the penalty for energy consumption. To assume that E_{s_i, s_j}^{fly} is the energy consumption of u_m flying from s_i to s_j , which could be calculated with **Definition 5**, and $E_m(s_j) = E_{s_j}^{col} + E_{s_j}^{com} + E_{s_j}^{hover}$ is the summation of energy consumption of data collection, data computation and hovering of u_m at s_j , which could be calculated with **Definitions 6 ~ 8**, $E_{s_i, s_j} = E_{s_i, s_j}^{fly} + E_m(s_j)$ could be taken as the total energy consumption caused by an action choice of u_m hovering at s_i . Accordingly, for either pair of two hovering points, but $s_i \neq s_j$, there is $E_{s_i, s_j} \neq 0$. If $s_i = s_j$, $E_{s_i, s_j} = 0$. All E_{s_i, s_j} could constitute a matrix of energy consumption, $\mathcal{M}(E) = \{E_{s_i, s_j}\}_{(K+1) \times (K+1)}, \forall (s_i, s_j) \in (S(o) \cup S(h))^2$, and for u_m hovering at $u_m(l)$, the negative reward as the penalty for energy consumption could be defined as

$$p_e(l) = -r_e \cdot \frac{E_m(l+1) - E_m(l) - \overline{\mathcal{M}(E)}}{\mathcal{M}(E)_{max}}, \quad (15)$$

where $\overline{\mathcal{M}(E)}$ is the mean of all elements in $\mathcal{M}(E)$, $\mathcal{M}(E)_{max}$ is the maximum in $\mathcal{M}(E)$, $r_e = 1$ is the penalty coefficient.

(2) To ensure all ground users to be fully served and the integrity of data collection and computation for ground users, if there is an unvisited hovering point while all UAVs being on the way departure station, the UAV would be given the penalty for incomplete coverage, which could be defined as

$$p_c(l) = -r_c \cdot \left(K - \sum_{i=1}^K \sum_{k=0}^K \sum_{m=1}^M x_{s_i s_k}^m \right), \quad (16)$$

where $r_c = 1$ is the penalty coefficient.

(3) If all UAVs could make a return trip without energy exhaustion, and the learned path is optimal for each UAV, the positive reward given to each UAV could be defined as

$$r(L) = 10^{1 + \frac{\mathcal{F}(L) - \mathcal{F}'(L)}{\mathcal{F}(L)}}, \quad (17)$$

where $\mathcal{F}(L)$ is the value of $\mathcal{P}1$ corresponding to the known optimal path, and $\mathcal{F}'(L)$ is the value of $\mathcal{P}1$ corresponding to the next path option.

B. Multi-Agent Cooperation for Service Fairness

To maximize the service fairness experienced by ground users through minimizing the maximum in minimum hop counts between ground users and their candidate serving UAVs, and the standard deviation of hop counts, it is necessary to reasonably optimize locations for both the departure station and hovering points to satisfy requirements of association and full coverage. Therefore, this problem could be transformed into a partially observable continuous MDP with multiple agents.

Proof: For the problem of location planning, the involved agents are $K + 1$ locations, which could be represented as $\Upsilon_s = \{s_0, s_1, \dots, s_K\}$, and agents could cooperate with each other to obtain the optimal $S(o) \cup S(h) = \{s_k(x_k, y_k, z_k)\}_{k \in \{0, \dots, K\}}$ while achieving the expected service fairness. At the time-slot t in successive intervals, for determining optimal locations, in terms of the state at the current location, each agent s_k would perform an action a_k

according to the continuous sequence decision mode, and generate a new location $s_k^{t+1}(x_k^{t+1}, y_k^{t+1}, z_k^{t+1})$. Therefore, the decision-making process could be considered as that the generation of location is related to the state at the current location, but has nothing to do with previous states, that is, $p(s_k^{t+1} | (s_k^t, a_k^t), \dots, (s_k^1, a_k^1)) = p(s_k^{t+1} | (s_k^t, a_k^t))$. To improve the efficiency of location planning while satisfying defined constraints, in each time-slot, UAVs hovering at agents could communication with each other to share the location information and the information of ground users to be served, $\mathcal{O}_{s_k}(t)$, but, sine the next actions adopted by agents are independent, the environment is partially observable for location planning. In conclusion, the location planning could be transformed into a partially observable continuous MDP. ■

Hence, a tuple $\langle \mathcal{S}_s, \mathcal{A}_s, \mathcal{O}_s, \mathcal{R}_s, \mathcal{P}_s, K \rangle$ is defined, $\mathcal{S}_s = \mathcal{S}_{s_0} \times \mathcal{S}_{s_1} \times \dots \times \mathcal{S}_{s_K}$ is the global state space, $\mathcal{A}_s = \mathcal{A}_{s_0} \times \mathcal{A}_{s_1} \times \dots \times \mathcal{A}_{s_K}$ is the action space shared by agents, $\mathcal{O}_s = \mathcal{O}_{s_0} \times \mathcal{O}_{s_1} \times \dots \times \mathcal{O}_{s_K}$ is the observation space and $\mathcal{O}_{s_k}$ is generated by s_k local information, $\mathcal{R}_s$ is the reward function, $\mathcal{P}_s : \mathcal{S}_s \times \mathcal{A}_s \rightarrow \mathcal{S}_s$ is the state transition probability function.

In the considered scenario, to better define the continuous MDP for location planning, the decision-making process would be conducted in consecutive time intervals during T , and the time-slot is ΔT . Accordingly, the number of time-slots is $\frac{T}{\Delta T}$, and the serial number of time-slot could be defined as $t \in \{1, 2, \dots, \frac{T}{\Delta T}\}$. In continuous steps, each agent could interact with the environment to improve the policy of location planning. As an illustration, at step t , each agent s_k could obtain the current observation $\mathcal{O}_{s_k}(t)$ from the global state space $\mathcal{S}_s(t) \triangleq \{\mathcal{O}_{s_k}(t) | \forall s_k \in \Upsilon_s\}$, $\mathcal{O}_{s_k}(t) : \mathcal{S}_s(t) \rightarrow \mathcal{O}_{s_k}(t)$, subsequently, with the observation, s_k could obtain rewards through executing corresponding actions, and the state would be transferred into a new one $\mathcal{S}_s(t+1)$. The whole process would be repeated until the trained optimal location planning is obtained.

To address the problem of achieving the service fairness among ground users through location planning, detail explanation of each component involved in the continuous MDP is stated as follows. Through comprehensively comparing popular policy optimization based multi-agent cooperative deep reinforcement learning approaches [35], [36], an algorithm based on the architecture of Actor-Critic is proposed as a promising solution, which is stated in Algorithm 2.

State: With the formulated optimization problem, the state space has five components, e.g., locations of the departure station and hovering points, locations of ground users, the communication radius of UAV, the communication radius of ground user, and the association between ground user and hovering point. The association could be determined with **Definition 4** and formulation (7). $\forall t \in \{1, 2, \dots, \frac{T}{\Delta T}\}$, the local state space, $\mathcal{S}_{s_k}(t) \in \mathcal{S}_{s_k}$, could be defined as $\mathcal{S}_{s_k}(t) = \{s_0, \dots, s_K, gu_1, \dots, gu_N, r_{gu}, r_{gu}^n, y_k^n\}$. The global state space shared by agents therefore could be further defined as $\mathcal{S}_s \triangleq \{\mathcal{S}_{s_k}(t)\}$.

Action: For s_k in the t -th time-slot, the chosen moving step size and direction, and shared information of locations constitute the action space, that is, $\mathcal{A}_{s_k}(t) = \{\zeta_{s_k}(t), \vartheta_{s_k}(t), s_0(t), \dots, s_K(t)\}$, where $\zeta_{s_k}(t)$ represents the

Algorithm 2 Multi-Agent Cooperative Actor-Critic for $\mathcal{P}2$ **Input:** Defined parameters.**Output:** Location planning.

```

1: With the number of agents,  $K + 1$ , generating initial
   locations for  $S(o) \cup S(h) = \{s_k^0(x_k^0, y_k^0, z_k^0)\}_{k \in \{0, \dots, K\}}$ ;
2: Initializing both the Actor network and the Critic network,
   and corresponding weights;
3: Obtaining the matrix  $\mathcal{M}(\text{hop})$  with Definition 4;
4: Obtaining the initial optimal value of  $\mathcal{P}2$ ,  $\mathcal{F}(k)$ ;
5: Initializing a  $Q$  table,  $\forall s \in \mathcal{S}_s$  &  $\forall a \in \mathcal{A}_s$ ;
   /* Updating the  $Q$  table and  $\mathcal{F}(k)$ . */
6: for  $k = 0$  to  $K$  do
7:   for  $t = 1$  to  $\frac{T}{\Delta T}$  do
8:     for  $s \in \mathcal{S}_s(t)$  do
9:       Selecting an action  $a$  ( $a \in \mathcal{A}_s$ ) with  $\varepsilon - greedy$ ;
10:      Calculating the corresponding possibility  $\pi(a|s)$ 
        with the strategy in the Actor network;
11:      Calculating the  $Q$  value with the strategy in the
        Critic network,  $Q(s, a)$ ;
        /* Calculating rewards. */
12:      if ( $\rho'(\text{cov.}) > \rho(\text{cov.})$ ) then
13:        Calculating the positive reward  $r_c(k)$ ;
14:      else
15:        Calculating the negative reward  $p_c(k)$ ;
16:      end if
17:      if ( $R_{\max} > R'_{\max}$ ) then
18:        Calculating the positive reward  $r_h(k)$ ;
19:      else
20:        Calculating the negative reward  $p_h(k)$ ;
21:      end if
22:      if ( $STD(R) > STD(R')$ ) then
23:        Calculating the positive reward  $r_d(k)$ ;
24:      else
25:        Calculating the negative reward  $p_d(k)$ ;
26:      end if
27:      Obtaining the instant reward,  $\mathcal{R}_s(s, a)$ , and the
        updated value of  $\mathcal{P}2$ ,  $\mathcal{F}'(k)$ ;
28:      if ( $\mathcal{F}'(k) < \mathcal{F}(k)$ ) then
29:         $\mathcal{F}(k) \leftarrow \mathcal{F}'(k)$ ;
30:      end if
31:      Updating the state after executing  $a$ ,  $s \rightarrow s'$ ;
32:      Calculating the  $Q$  value with the strategy in the
        Critic network,  $Q(s', a)$ ;
33:      Calculating the Temporal Difference,  $\delta =$ 
         $\mathcal{R}_s(s, a) + \gamma \cdot Q(s', a) - Q(s, a)$ ;
34:      Updating weights in the Critic network;
35:      Calculating the loss function in the Actor network,
         $F_{loss} = -\delta \cdot \log \pi(a|s)$ ;
36:      Updating weights in the Actor network;
37:      if (convergence) then
38:        Determining the location for  $s_k(x_k, y_k, z_k)$  with
         $\mathcal{F}(k)$ ;
39:      else
40:         $s \leftarrow s'$ , repeat steps: 9 ~ 36;
41:      end if
42:    end for
43:  end for
44: end for

```

moving step size and $\vartheta_{s_k}(t)$ ($\vartheta_{s_k}(t) \in [0, 2\pi]$) represents the moving direction. Therefore, the global action space shared by agents could be further defined as $\mathcal{A}_s \triangleq \{\mathcal{A}_{s_k}(t)\}$. However, to ensure each ground user to be associated with only one hovering point of UAV, *Constraint 3* should be satisfied in the global action space.

Observation: Assuming that agents could communicate with each other to obtain locations of hovering points and associations between ground users and hovering points, therefore, for s_k in the t -th time-slot, the local observation space $\mathcal{O}_{s_k}(t) \in \mathcal{O}_{s_k}$ ($\forall t \in \{1, 2, \dots, \frac{T}{\Delta T}\}$) could be defined as $\mathcal{O}_{s_k}(t) = \{s_0, \dots, s_K, y_k^n\}$, and the global observation space could be further defined as $\mathcal{O}_s \triangleq \{\mathcal{O}_{s_k}(t)\}$.

Reward: To improve the search efficiency of location planning while satisfying defined constraints, and obtain optimal location planning for the departure station and hovering points of UAVs, negative and positive rewards could be introduced from perspectives of the full coverage that serving areas of hovering points should cover all ground users, the reduction of the maximum in minimum hop counts between ground users and their candidate serving UAVs, and the reduction of the standard deviation in minimum hop counts.

(1) To enable the full coverage with *Constraint 4* that serving areas of determined hovering points could cover all ground users, for each action choice while the UAV flying into the service coverage area of the k -th hovering point, with the assumption that the reward has a linear relationship with the percentage increase in coverage rate, if the achieved coverage rate of ground users could be increased, the agent would be given a positive reward defined as

$$r_c(k) = r_{cov.}^{pos.} \cdot \frac{\rho'(\text{cov.}) - \rho(\text{cov.})}{\rho(\text{cov.})}, \quad (18)$$

where $r_{cov.}^{pos.} = 1$ is the reward coefficient, $\rho'(\text{cov.})$ is the coverage rate achieved with the current action choice, and $\rho(\text{cov.})$ is the historically optimal coverage rate achieved with action choices. Otherwise, if there are uncovered ground users, the agent would be given a penalty for incomplete coverage, which could be defined as

$$p_c(k) = -r_{cov.}^{neg.} \cdot N(\text{uncov.}), \quad (19)$$

where $r_{cov.}^{neg.} = 1$ is the penalty coefficient, and $N(\text{uncov.})$ is the number of uncovered ground users.

(2) With $\mathcal{P}2$, to ensure the reduction of the maximum in minimum hop counts between ground users and their candidate serving UAVs, for each action choice while the UAV flying into the service coverage area of the k -th hovering point, with the assumption that the reward has a linear relationship with the percentage reduction of maximum in minimum hop counts, if the maximum in minimum hop counts could be declined, the agent would be given a positive reward defined as

$$r_h(k) = r_{hop}^{pos.} \cdot \frac{R_{\max} - R'_{\max}}{R_{\max}}, \quad (20)$$

where $r_{hop}^{pos.} = 1$ is the reward coefficient, $R'_{\max}$ is the maximum in minimum hop counts achieved with the current action choice, $R_{\max}$ is the historically optimal maximum in minimum hop counts achieved with action choices. Otherwise,

TABLE II
CONFIGURATIONS OF SIMULATION PARAMETERS

Parameters	Values	Parameters	Values
r_{gu} (Communication radius of gu)	1.5 m	r_u (Communication radius of UAV)	4 m
α (Factor introduced in $\mathcal{P}1$)	0.5	β (Factor introduced in $\mathcal{P}2$)	0.5
E_m^{full} (Full energy capacity of u_m)	200 J	B_m (Uplink frequency bandwidth)	20 MHz
P_k^i (Uplink transmission power of gu_k^i)	0.02 W	g_k^i (Uplink channel gain of gu_k^i)	2 dB
θ (Noise power spectral density)	-124.29 dBm/Hz	δ (Energy consumption coefficient of UAV flight)	1
κ_m (Energy coefficient of u_m)	1×10^{-9}	c_m (Computing capacity of u_m)	1 GHz
D_n (Data volume of gu_n)	[0,100] KB	ξ (Energy consumption coefficient of UAV hovering)	1

with the assumption that the penalty has a linear relationship with the percentage increase of maximum in minimum hop counts, the agent would given a penalty for violating $\mathcal{P}2$, which could be defined as

$$p_h(k) = -r_{hop}^{neg.} \cdot \frac{R'_{\max} - R_{\max}}{R'_{\max}}, \quad (21)$$

where $r_{hop}^{neg.} = 1$ is the penalty coefficient.

(3) With $\mathcal{P}2$, to ensure the reduction of the standard deviation in minimum hop counts, for each action choice while the UAV flying into the service coverage area of the k -th hovering point, with the assumption that the reward has a linear relationship with the percentage reduction of standard deviation, if the standard deviation in minimum hop counts could be declined, the agent would be given a positive reward defined as

$$r_d(k) = r_{dev.}^{pos.} \cdot \frac{STD(R) - STD(R)'}{STD(R)}, \quad (22)$$

where $r_{dev.}^{pos.} = 1$ is the reward coefficient, $STD(R)'$ is the standard deviations achieved with the current action choice and $STD(R)$ is the historically optimal standard deviation achieved with action choices. Otherwise, assuming the penalty has a linear relationship with the percentage increase of standard deviation, the agent would given a penalty for violating $\mathcal{P}2$, which could be defined as

$$p_d(k) = r_{dev.}^{neg.} \cdot \frac{STD(R)' - STD(R)}{STD(R)'}, \quad (23)$$

where $r_{dev.}^{neg.} = 1$ is the penalty coefficient.

VI. NUMERICAL RESULTS

A. Simulation Configurations

In this paper, due to the cost of large-scale deployment and implementation, simulation experiments are conducted for effectively verifying the overall performance of investigations. Python 3.9.0 is used as the programming tool, and experiments are running on the server: Ubuntu 18.04.6 LTS operating system, Intel(R) Xeon(R) Silver 4210R CPU at 2.40GHz, and 64 GB of RAM. Even so, to fully demonstrate the practical feasibility and effectiveness of those investigations, as shown in Fig. 3, we also comprehensively consider attributes of involved entities in a real-world hardware platform, and further configure simulation parameters in Table II.



Configuration	Product Model
UAV	DJI MATRICE 100
Motor	DJI 3510
Electrically controlled model	DJI E SERIES 620D
Propeller	DJI 1345s
Flying control system	N1
Charger	A14-100P1A
Remote-controller	C1
Battery	TB47D

(a) UAV

Configurations	Product model
Development Board	STM32F407VET6
Wireless Module	(SIM900A) GSM/GPRS
Sensors	(DHT11) Temperature & Humidity sensor
	(SDS011) Laser PM2.5 sensor
	(ZYMQ-7) Carbon monoxide sensor
	(TaoTimeClub) Sound sensor
	(MQ131) Ozone gas detection sensor
Data gathering terminal	(MS1100-P111) VOC gas sensor

(b) Data collection terminal of ground user

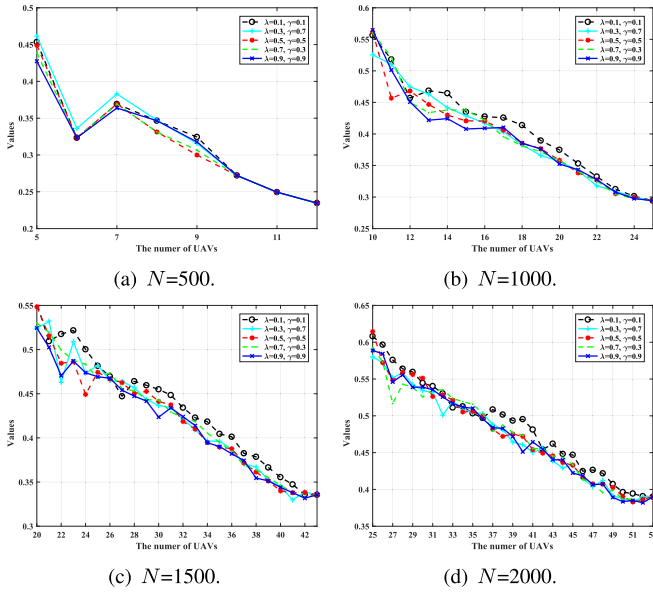
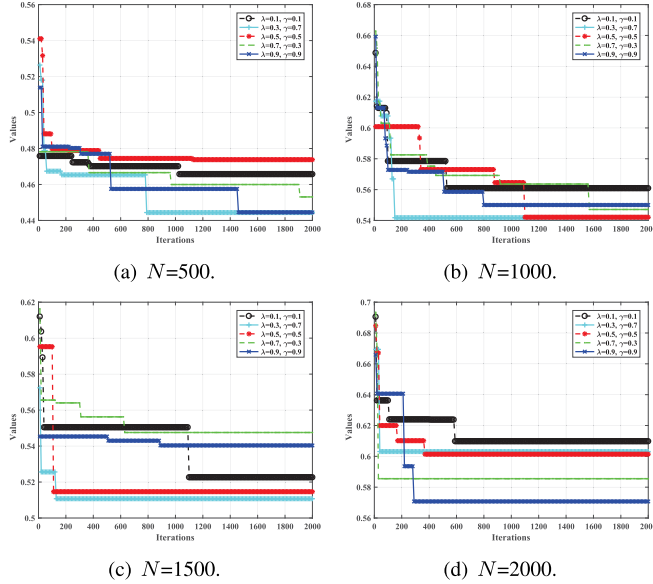
Fig. 3. Attributes of involved entities in a real-world platform.

B. Evaluation of Energy Efficiency

Here, to effectively verify the achievement of the desired optimization objective of energy efficiency through reasonable trajectory planning, effectiveness and convergence of the proposed Algorithm 1 for $\mathcal{P}1$ are evaluated.

Given the number of ground users, variations of the value of $\mathcal{P}1$ with the number of UAVs are shown in Fig. 4. In the proposed Algorithm 1, the exploring rate used in Q-learning is adaptive to the number of iterations, therefore, different configurations for the combination of the learning rate λ and the discount rate γ are used to evaluate the achieved effectiveness of Algorithm 1. From Fig. 4, we could clearly see that, along with the increase of the number of ground users, the number of deployed UAVs required to achieve the full converge is increasing as well. Nevertheless, no matter how the combination of λ and γ is configured, the value of $\mathcal{P}1$ is declining through conducting reasonable trajectory planning, which indicates variations of the value of $\mathcal{P}1$ are consistent with the desired optimization objective of energy efficiency, and the effectiveness of Algorithm 1 is well demonstrated.

Given the number of ground users, through deploying the optimal number of UAVs to achieve the full converge,

Fig. 4. Variations of the value of $\mathcal{P}1$ with the number of UAVs.Fig. 5. Variations of the value of $\mathcal{P}1$ with the number of iterations.

variations of the value of $\mathcal{P}1$ with the number of iterations are shown in Fig. 5. Similarly, different configurations for the combination of λ and γ are used to evaluate the achieved convergence of Algorithm 1. From Fig. 5, we could clearly see that, no matter how the combination of λ and γ is configured, along with the increase of the number of iterations, the convergence of the value of $\mathcal{P}1$ could be well achieved, which indicates that, the convergence of Algorithm 1 is well demonstrated, as well as, with the constraint of limited onboard battery capacities of UAVs, the desired optimization objective of energy efficiency could be well achieved through conducting reasonable trajectory planning.

Given the number of ground users, with the optimal number of deployed UAVs determined by Algorithm 1, the trajectory planning of UAVs is shown in Fig. 6, and we could clearly see that, with the determined optimal number of deployed UAVs,

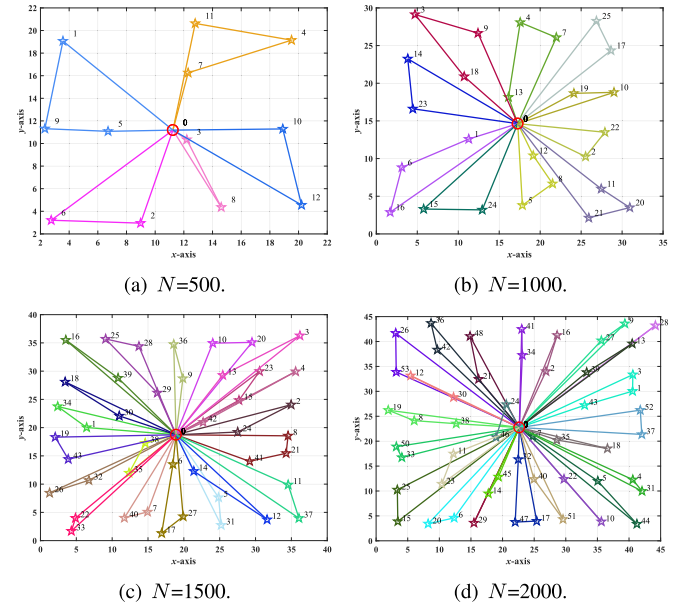


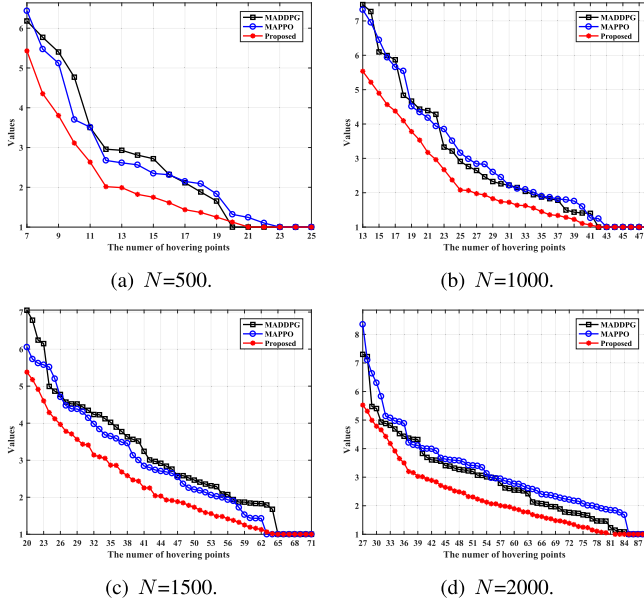
Fig. 6. Trajectory planning of UAVs.

the full coverage is well achieved through ensuring the serving coverage area of each determined hovering point to be visited by only one UAV, and each UAV to make only one return trip for providing complete service, which further indicates that, the effectiveness of Algorithm 1 is well demonstrated, and the desired optimization objective of energy efficiency is well achieved with defined constraints.

C. Evaluation of Service Fairness

Here, to effectively verify the achievement of the desired optimization objective of service fairness through conducting reasonable location planning for the departure station and hovering points, effectiveness and convergence of the proposed Algorithm 2 for $\mathcal{P}2$ are evaluated, and the two algorithms, multi-agent proximal policy optimization (MAPPO) [37] and multi-agent deep deterministic policy gradient (MADDPG) [38], are used for comparative studies.

- **MAPPO:** Within the framework of Centralized Training with Decentralized Execution (CTDE), MAPPO could use an Actor network and a Critic network to train multiple agents in a centralized manner, and utilize the global information of all agents during the training phase, however, in the execution phase, each agent could only select an action in terms of the local observations, and the achieved performance might be degraded due to the incomplete observations.
- **MADDPG:** Within the framework of CTDE, MADDPG uses multiple Actor networks and Critic networks to train multiple agents. However, it is necessary to take longer time to perform the training, and the convergence rate is relatively slow. Additionally, without being influenced by other agents, each agent separately has a Actor network and a Critic network to perform the training operation in terms of the local policy. However, due to being lack of cooperation among agents, the policy of each agent would be prone to the local optimality, which inversely influences the global performance of the algorithm.

Fig. 7. Variations of the value of $\mathcal{P}2$ with the number of hovering points.

Given the number of ground users, variations of the value of $\mathcal{P}2$ with the number of hovering points are shown in Fig. 7, and we could clearly see that, along with the increase of the number of ground users, the number of determined hovering points required to achieve the full converge is increasing as well. Nevertheless, for the three comparative algorithms, the obtained value of $\mathcal{P}2$ is declining, which indicates that, variations of the value of $\mathcal{P}2$ are consistent with the desired optimization objective of service fairness, and multi-agent deep reinforcement learning algorithms are suitable for solving the problem of defined continuous MDP. For the proposed Algorithm 2, in each time-slot, with well designed reward functions, the Critic network and the Actor network could cooperatively update involved weights through separately calculating the Temporal Difference, $\delta = \mathcal{R}_s(s, a) + \gamma \cdot Q(s', a) - Q(s, a)$, and the loss function, $F_{loss} = -\delta \cdot \log \pi(a|s)$, maximizing cumulative rewards of long-term expectations therefore could be achieved. Accordingly, the obtained value of $\mathcal{P}2$ is the best in them obtained by the three comparative algorithms, which indicates that the effectiveness of Algorithm 2 is well demonstrated.

Given the number of ground users, through deploying the optimal number of hovering points to achieve the full converge, variations of the value of $\mathcal{P}2$ with the number of iterations are shown in Fig. 8. For the three comparative algorithms, due to conducting the cooperation between the Critic network and the Actor network to find the optimal solution in Algorithm 2, there is a faster convergence of the value of $\mathcal{P}2$, which indicates that, the convergence of Algorithm 2 is well demonstrated, as well as, with the constraint of limited onboard battery capacities of UAVs, the desired optimization objective of service fairness could be well achieved through conducting reasonable location planning for hovering points.

Given the number of ground users, with the optimal number of hovering points determined by Algorithm 2, the location planning of hovering points is shown in Fig. 9, and we could clearly see that, with the determined optimal number

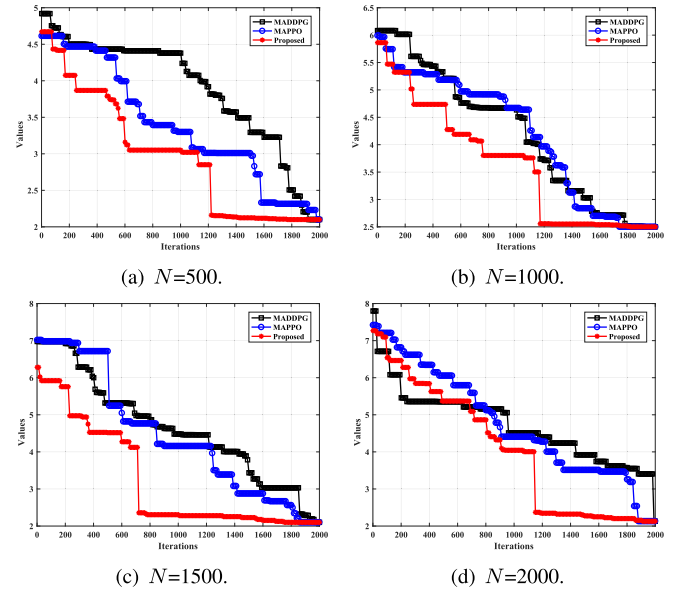
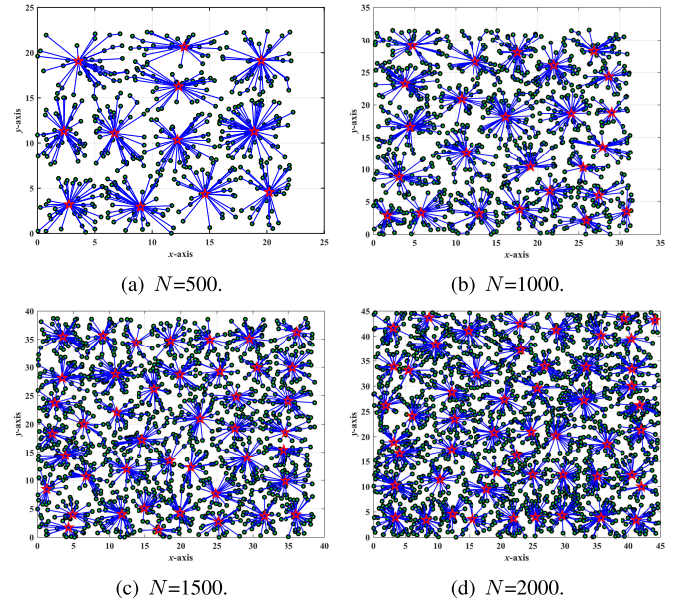
Fig. 8. Variations of the value of $\mathcal{P}2$ with the number of iterations.

Fig. 9. Location planning of hovering points.

of hovering points, the full coverage is well achieved through ensuring the serving coverage area of each determined hovering point to be visited by only one UAV, and each ground user to be only associated with a hovering point of UAV, which further indicates that, the effectiveness of Algorithm 2 is well demonstrated, and the desired optimization objective of service fairness is well achieved with defined constraints.

VII. CONCLUSION AND FUTURE WORK

In UAVs empowered aerial computing systems, to improve the energy efficiency of UAVs and the service fairness experienced by ground users while achieving reasonable computing power scheduling, a joint optimization with constraints is formulated in this paper. Through considering complex coupling associations among the departure station, flight paths and

hovering points, the trajectory planning of UAVs is conducted to maximize the energy efficiency, and the location planning for both the departure station and hovering points is conducted to maximize the service fairness. As promising solutions, investigated problems are firstly proven to be Markov decision processes, on the basis, a multi-agent cooperative Q-learning approach is proposed for optimizing trajectory planning while maximizing the energy efficiency, and a policy optimization based multi-agent cooperative deep reinforcement learning approach using the architecture of Actor-Critic is proposed for optimizing location planning while maximizing the service fairness. Comparative experiments show that the performance achieved by investigations has obvious advantages.

With the development of emerging intelligent services in 6G mobile systems, to satisfy multi-dimensional extreme performance requirements, it is necessary to deeply integrate the wireless sensing function with the wireless transmission function, and use the ubiquitous computing power for auxiliary computation. In this context, UAVs empowered aerial computing systems would be a powerful complement for the integration of sensing, communication, and computing for edge intelligence in 6G networks, therefore, formulating more complete optimization models with more comprehensive considerations of inherent limitations of UAVs and diverse requirements of ground users, and comparing more versions of multi-agent reinforcement learning algorithms, are still interests for conducting further explores. Especially, the relevant research progress would be migrated and applied into integrated Ground-Air-Space wireless networks for 6G mobile.

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